Parul® University

Cyber Security

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DIGITAL LEARNING CONTENT

CHAPTER-3

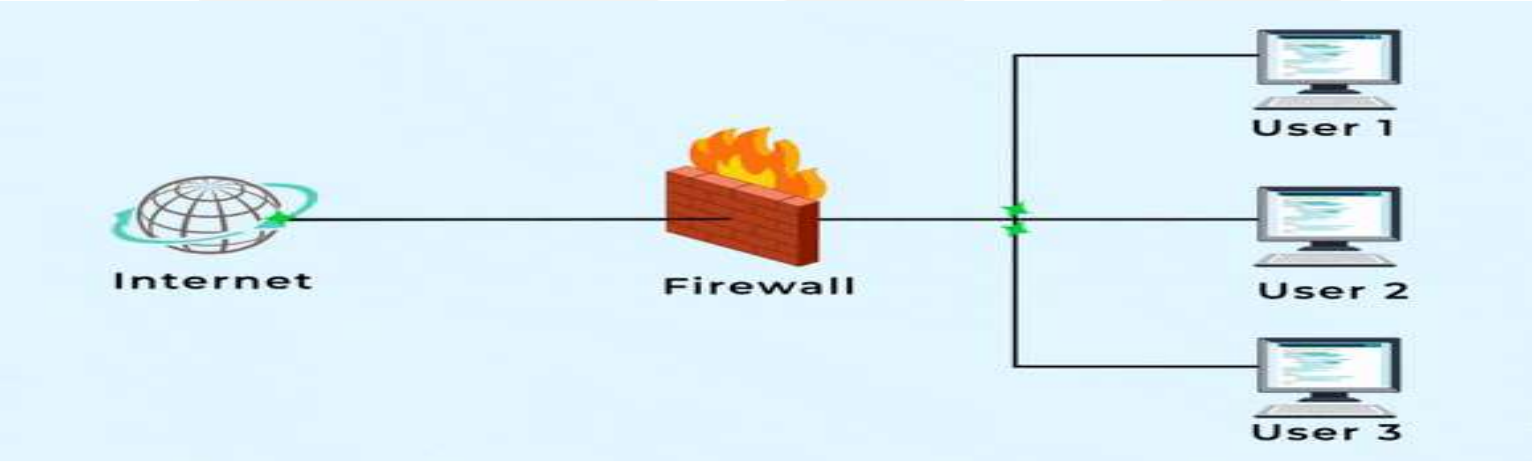
Network Defense tools

Firewalls Basics

A **firewall** is a device which is used to **control** the **flow of traffic** into and out-of network. In other words, it is a **security device** which **installed between** two networks, **internal network** to **outside network** (more often the internet).

Based on the **rule define** in the firewall **data will be passed** to one network to other network.

The primary job of a firewall is to **secure the inside network from the internet**. Systems on one side of the firewall are protected from systems on the other side.



Cont…

For example:

Consider **LAN** is corporate or our campus network and **WAN** is internet.

If we place firewall between the two networks then it will control the flow of the whole traffic and based on rule define into firewall.

It will **allow or deny** the traffic.

Firewalls generally **filter** traffic based on **two methodologies**:

1. A firewall can **allow any traffic** except what is specified as restricted part. It depends on the type of firewall used, the source, the destination addresses, and the ports.
2. A firewall can **deny any traffic** that does not meet the specific criteria based on the network layer on which the firewall operates.

Cont…

Source: TechTarget: Comparing inbound and outbound firewalls

Firewall Types

- Firewall is the first destination for the traffic coming to your internal network.
- So, **anything** which **comes** to your **internal network passes through** the **firewall** and any outgoing traffic will also pass through the firewall before leaving your network completely.
- This is the reason that sometimes this type of firewall filter is also called **screening routers**.
- Firewall types the way a firewall provides greater protection relies on the firewall itself, and on the policies that are configured on it.
- The Following types of firewall are:

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1. Packet-Filter Firewall
2. Circuit-Level Gateways
3. Stateful Packet-Inspection (SPI)
4. Proxy Firewall
5. Application Gateways
6. Next-Gen firewalls
7. Software Firewall
8. Hardware Firewall
9. Cloud Firewall

Packet Filtering Firewall

- As the most "basic" and oldest type of firewall architecture, **Packet-filtering** firewalls basically **create** a **checkpoint** at a traffic router or switch.
- The firewall performs a simple **check** of the **data packets** coming through the router—inspecting information such as the **destination** and **origination IP address**, **packet type**, **port number**, and other surface-level information **without opening** up the **packet** to inspect its contents.
- If the information **packet doesn't pass** the **inspection**, it is **dropped**.
- The **good thing** about these firewalls is that they **aren't very resource-intensive**.
- This means they don't have a **huge impact** on **system performance** and are **relatively simple**.
- However, they're also **relatively easy** to **bypass** compared to firewalls with more robust inspection capabilities.

Proxy Firewalls

- **Proxy** firewalls operate at the **application layer** to **filter incoming traffic** between your network and the traffic source—hence, the name "**application-level gateway**."
- Rather than letting traffic connect directly, the proxy firewall first establishes a connection to the source of the traffic and **inspects** the **incoming data packet**.
- This check is similar to the stateful inspection firewall in that it looks at both the packet and at the TCP handshake protocol.
- However, proxy **firewalls** may also **perform deep-layer packet inspections**, **checking** the **actual contents** of the information packet to **verify** that it contains **no malware**.

Cont…

- Once the check is complete, and the packet is approved to connect to the destination, the proxy sends it off.
- This creates an extra layer of separation between the “client” (the system where the packet originated) and the individual devices on your network—obscuring them to **create additional anonymity** and **protection** for your network.
- It’s that they can create **significant slowdown** because of the extra steps.

Circuit-Level Gateways

- As another simplistic firewall type that is meant to quickly and easily approve or deny traffic without consuming significant computing resources.
- Circuit-level gateways work by **verifying** the **transmission control protocol** (TCP) **handshake**.
- This TCP handshake check is designed to **make sure** that the session the **packet** is **from legitimate**.
- While **extremely resource-efficient**, these firewalls **does** not **check the packet** itself.
- So, if a **packet held malware**, but had the right TCP handshake, it **would pass** right through.
- This is why circuit-level gateways are **not enough** to **protect** your **business** by themselves.

Stateful Inspection Firewalls

- These firewalls **combine both packet inspection technology** and **TCP handshake** verification to create a level of protection greater than either of the previous two architectures could provide alone.

- However, these firewalls do put **more** of a strain on **computing resources** as well. This may **slow down** the **transfer** of legitimate packets compared to the other solutions.

Application Level Firewall

- These firewalls operate at the application level.
- In other words, they filter the **traffic only** with regards to the **application** (or **service**) for which they are intended.
- For example, a firewall for monitoring traffic to all the web applications your network uses.

Next-Generation Firewalls

- Many of the **most recently-released** firewall products are being advertised as "**next-generation**" architectures.
- Some common features of next-generation firewall architectures **include deep-packet inspection** (checking the actual contents of the data packet), **TCP handshake checks**, and **surface-level packet inspection**.
- Next-generation firewalls may include other technologies as well, such as **intrusion prevention systems** (**IPSs**) that work to **automatically stop attacks** against your network.
- The issue is that there is no one definition of a next-generation firewall, so it's important to verify what specific capabilities such firewalls have before any conclusion.

Software Firewalls

- **Software firewalls** include any type of firewall that is **installed** on a **local device rather than** a **separate** piece of **hardware**.
- The big benefit of a software firewall is that it's highly useful for creating defense in depth by **isolating individual network endpoints** from one another.
- However, **maintaining** individual software firewalls on different devices can be **difficult** and **time-consuming**.
- Furthermore, not every device on a network may be compatible with a single software firewall, which may mean having to use several different software firewalls to cover every asset.

Hardware Firewalls

- **Hardware firewalls** use a **physical appliance** that acts in a manner similar to a traffic router to **intercept data packets** and **traffic requests** before they're connected to the network's servers.

- Physical appliance-based firewalls like this excel at perimeter security by **making sure malicious traffic** from outside the network is **stopped before** the company's **network endpoints** are **exposed** to risk.

- The actual **capabilities** of a hardware firewall **may vary depending** on the **manufacturer—some** may have a more limited capacity to handle simultaneous connections than others.

Cloud Firewalls

- Whenever a cloud solution is used to deliver a firewall, it can be called a cloud firewall, or **firewall-as-a-service** (**FaaS**).
- **Cloud** firewalls are considered **synonymous** with **proxy** firewalls by many, since a cloud server is often used in a proxy firewall setup.
- The **big benefit** of having cloud-based firewalls is that they are **very easy** to **scale** with your organization. As your needs grow, you can **add additional capacity** to the cloud server to filter larger traffic loads.
- Cloud firewalls, like hardware firewalls, excel at perimeter security.

Firewall vs Packet Filters

- A firewall is a computer connected to both a private (protected) network and a public (unprotected) network, which receives and resubmits specific kinds of network requests on behalf of network clients on either the private or public network.
- **Firewalls involve proxies**. A proxy **acts as** a **middle-man** in a network transaction.
- Rather than allowing a client to speak directly to a server, the proxy server receives the request from the client, and then resubmits the request, on behalf of the client, to the target server.

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- Each protocol or type of network transaction typically requires its own proxy program, and an administrator enables or installs specific proxies to determine what kinds of services will be allowed between the two networks.
- **Firewalls are not routers** or address translators.
- The internal network uses private address space. **Neither side of** the **firewall knows** about the **address space on** the **other side** of the firewall, and does not know how to route data to the other side of the firewall.

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- A **packet filter** is a **set of rules**, applied to a stream of data packets, which is used to **decide** whether to **permit** or **deny** the **forwarding** of each **packet**.
- These rules are usually on a router or in the routing layer of a computer's network protocol stack.
- Using a packet filter, an administrator can **dictate what types** of **packets** are **allowed** into or out of a network or computer.
- **Prevents** the **outside network** from having knowledge of the address space on the protected network.
- However, aside from translating the addresses of the internal network, **packets are forwarded as received** through the unit, and **no proxies** are **involved**.
- Any **good firewall** will also **employ packet filtering**.
- This is done to protect the firewall itself from intrusion and to isolate intruders from the internal network.

Packet Characteristic to Filter

- By using Packet filtering, firewall will create rule and based on rule it will allow or block incoming packet.
- Most firewalls and packet filters have the ability to examine the following characteristics of network traffic:

- **Type of protocol (IP, TCP, UDP, ICMP, IPsec, etc.)**
- **Source IP address and port**
- **Destination IP addresses and port**
- **ICMP message type**
- **TCP flags (ACK, FIN, SYN, etc.)**
- **Network interface on which the packet arrives**

Cont…

For example, if you wanted to **block incoming ping packets** (ICMP echo requests) to your home network of 192.168.1.0/24, you could write something like the following rule.

The important components of the rule are the action (deny), the packet attributes (ICMP protocol, specifically “ping” types), the direction of the rule (packets “from” one source “to” another), and the type of source (a network address range like 192.168.1.0/24).

deny proto icmp type 8:0 from any to 192.168.1.0/24

Other way that is if you wanted to allow incoming web traffic to 192.168.1.50 but deny everything else, you would create two rules. The first one would specify the direction of web traffic to a specific TCP ***port*** on a specific host. The second one would make sure all other traffic is denied. Those rules would look like the following:

For allow: allow proto tcp from any to 192.168.1.50:80

For block: deny proto all from any to 192.168.1.0/24

Stateless Firewalls

- If the information about the **passing packets is not remembered** by the firewall, then this type of filtering is called **stateless packet filtering**.
- **These types** of firewalls **are not smart enough** and can be **fooled** very **easily by** the **hackers**.
- These are especially **dangerous for UDP** type of data packets.
- The reason is that, the allow/deny decisions are taken on packet by packet basis and these are not related to the previous allowed/denied packets

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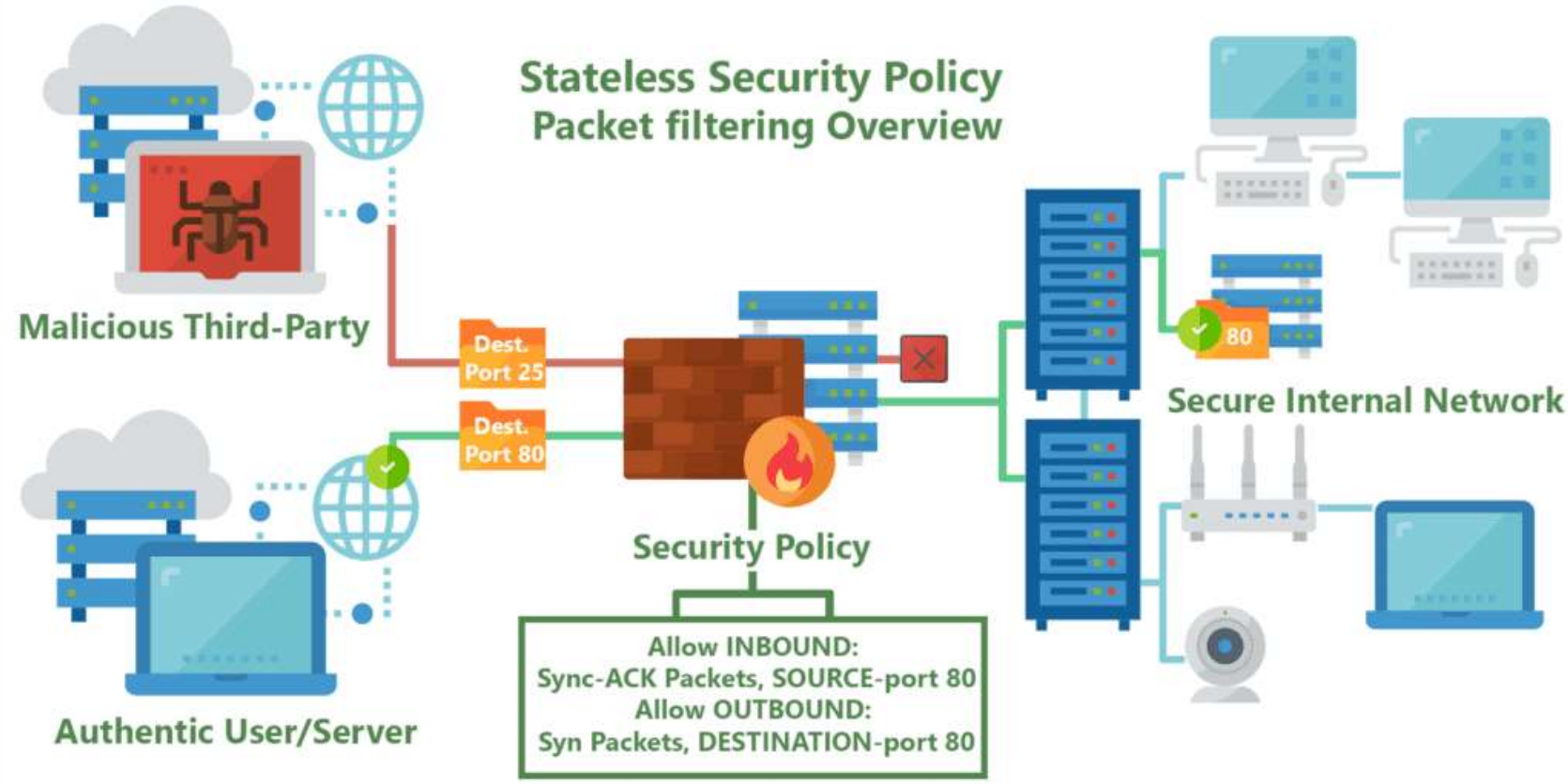
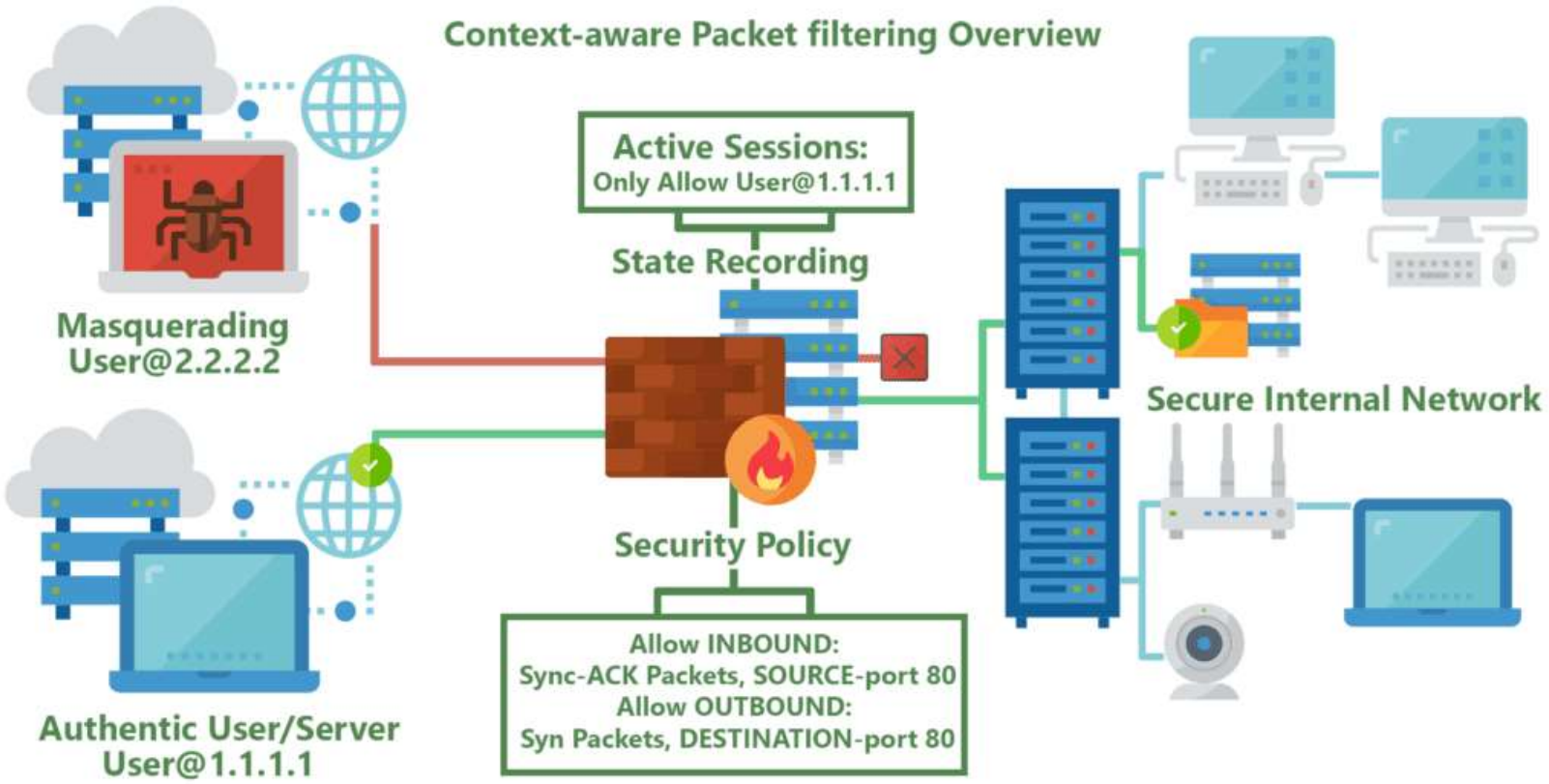

Stateless Security Policy
Packet filtering Overview
Malicious Third-Party
Dest. Port 25
Dest. Port 80
Authentic User/Server
Security Policy
Allow INBOUND:
Sync-ACK Packets, SOURCE-port 80
Allow OUTBOUND:
Syn Packets, DESTINATION-port 80
80
Secure Internal Network

Stateful Firewalls

- If the firewall **remembers** the **information about** the **previously passed packets**, then that type of filtering is **Stateful packet filtering**.
- These can be termed as **smart firewalls**. This type of **filtering** is also **known** as **Dynamic packet filtering**.

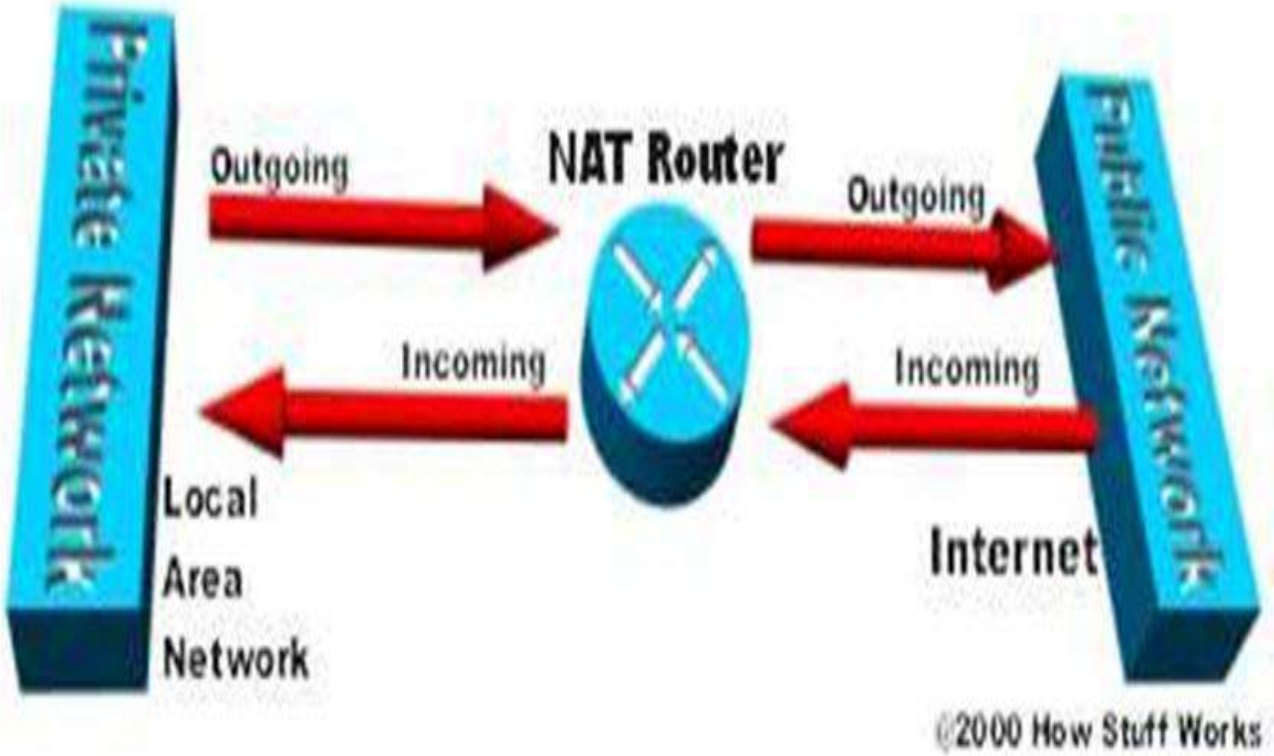
Cont…


Context-aware Packet filtering Overview
Active Sessions:
Only Allow User@1.1.1.1
State Recording
Masquerading
User@2.2.2.2
Secure Internal Network
Security Policy
Allow INBOUND:
Sync-ACK Packets, SOURCE-port 80
Allow OUTBOUND:
Syn Packets, DESTINATION-port 80
Authentic User/Server
User@1.1.1.1

Network Address Translation (NAT)

- Network Address Translation (NAT) is **designed for IP** address **conservation**.
- **Network Address Translation** (NAT) is method of **connecting multiple computers** to the Internet **using one IP address**.
- It enables **private IP networks** that use unregistered IP addresses to **connect to the Internet**.
- **NAT operates on** a **router**, usually connecting two networks together, and **translates** the **private addresses** in the internal network **into legal addresses**, **before** packets are **forwarded to another network**.
- Here in figure we see that NAT router operate between internal networks to Public network. By using NAT router when traffic come from the private network after that NAT router convert private network IP to other IP before transfer packet to another network.

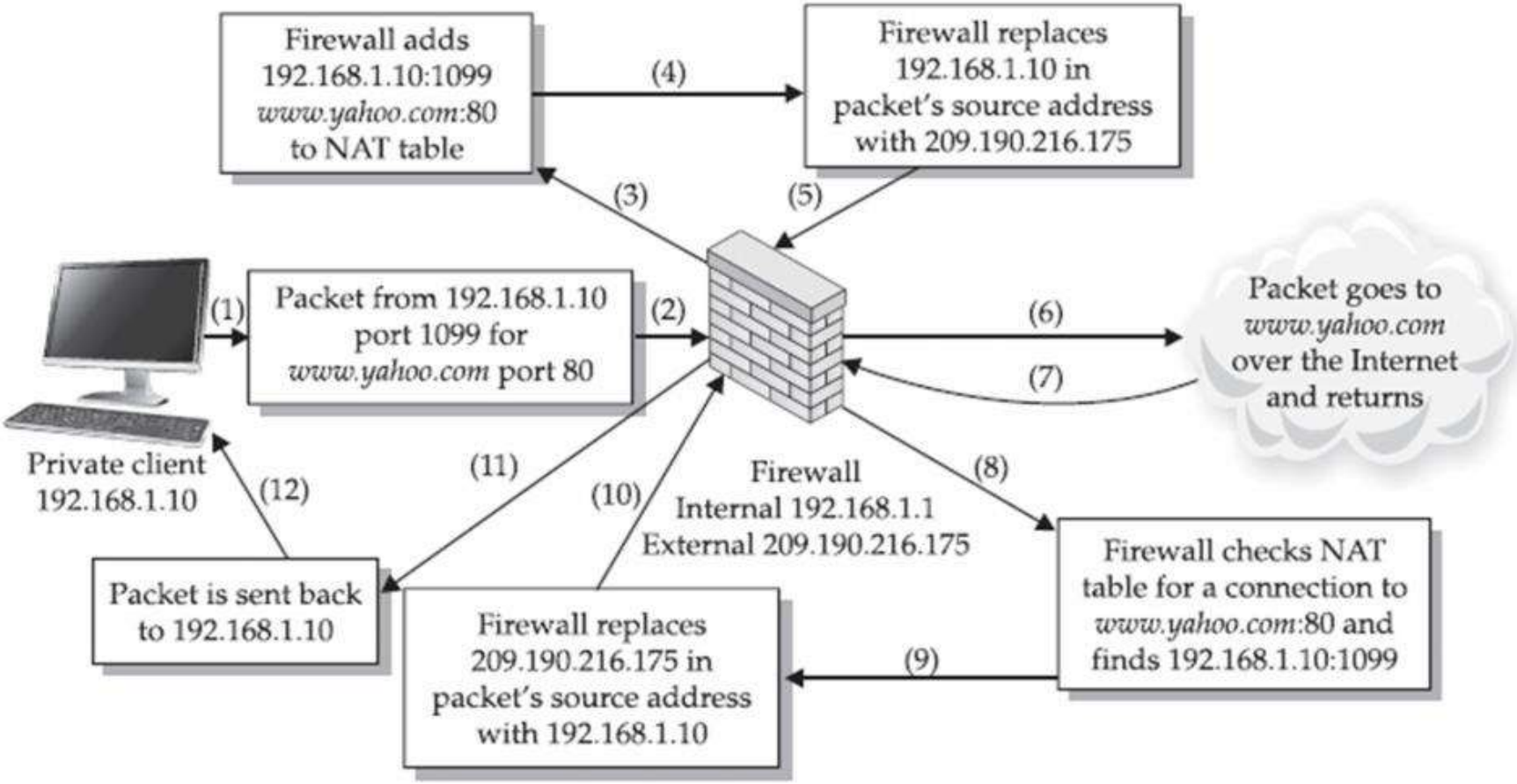
Cont…


Private Network
Outgoing
NAT Router
Outgoing
Public Network
Incoming
Incoming
Local
Area
Network
Internet
©2000 How Stuff Works

Cont…

- Here in below figure, we can easily understand flow of each step one by one.
- Here we see that **Firewall** that work **as NAT device** for IP converting and transfer packet to other network.

Cont…


Firewall adds 192.168.1.10:1099 www.yahoo.com:80 to NAT table
(4)
Firewall replaces 192.168.1.10 in packet's source address with 209.190.216.175
(3)
(5)
(1)
Packet from 192.168.1.10 port 1099 for www.yahoo.com port 80
(2)
(6)
(7)
Packet goes to www.yahoo.com over the Internet and returns
Private client 192.168.1.10
(12)
(11)
(10)
Firewall Internal 192.168.1.1 External 209.190.216.175
(8)
Packet is sent back to 192.168.1.10
Firewall replaces 209.190.216.175 in packet's source address with 192.168.1.10
(9)
Firewall checks NAT table for a connection to www.yahoo.com:80 and finds 192.168.1.10:1099

Cont…

- NAT maps Private IPs to Public IPs.
- A short term solution to the problem of the depletion of IP addresses
- It is required because of shortage of IPv4 Address.
- Whatever connects directly into Internet must have public (globally unique) IP address
- So Private IP addresses can be used within a private network
- Three address ranges are reserved for private usage
- 10.0.0.0/8
- 172.16.0.0/16 to 172.31.0.0/16
- 192.168.0.0/24 to 192.168.255.0/24

- A private IP is mapped to a Public IP, when the machine has to access the Internet

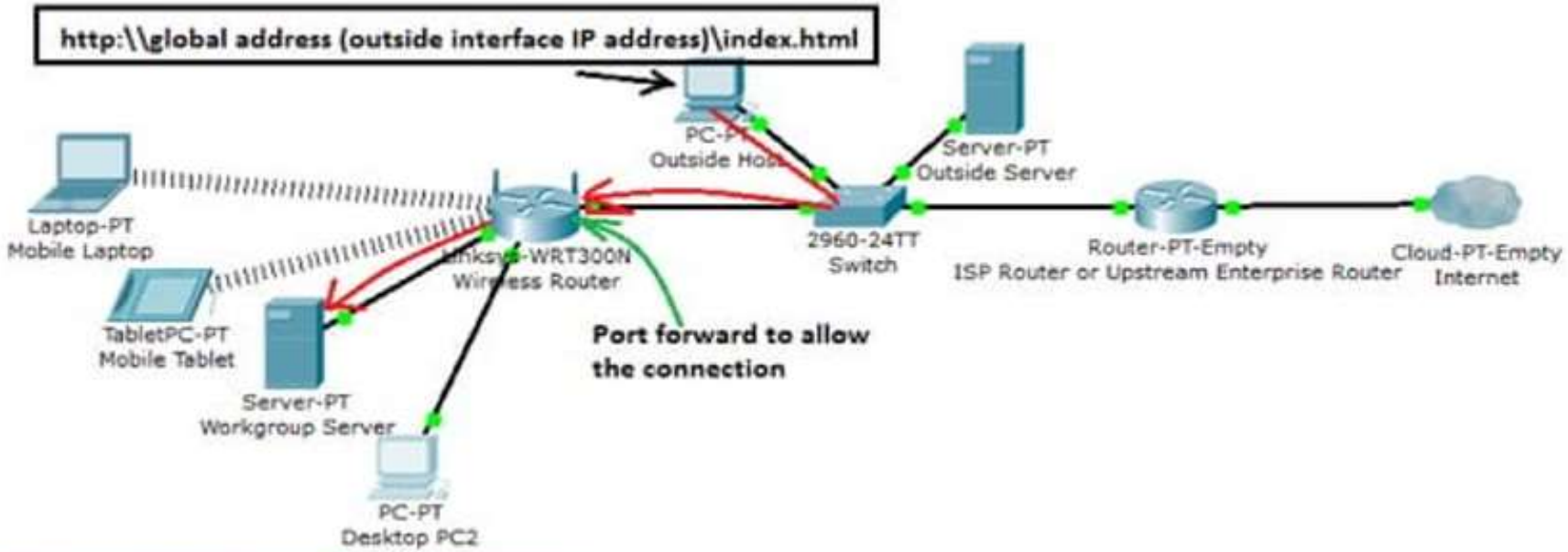
Names	Meaning
Inside Local	Source host inside address before translation typically an RFC 1918 address
Outside Local	Address from which source host is known on the Internet. This is usually the address of the router interface connected to ISP—the actual Internet address.
Inside Global	Source host address used after translation to get onto the Internet. This is also the actual Internet address.
Outside Global	Address of outside destination host. The real Internet address.

Port Forwarding

- **Port forwarding** or **port mapping** is an application of network address translation (NAT) that redirects a communication request from **one address and port number combination to another** while the packets are crossing a network gateway, such as a router or firewall.
- Port **forwarding allows computers of different network** (Internet) **to connect** to a **specific computer** or **service** within a **private local-area network** (LAN).
- Ports can be "opened" and "closed" in the firewall, which determines which types of traffic are allowed in or out.

Port Forwarding

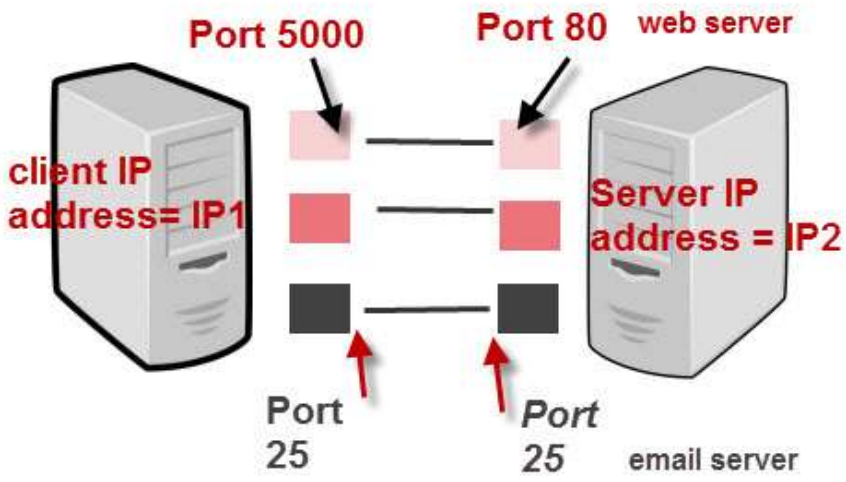
- Port forwarding allows remote hosts from the Internet or outside network to connect to a server on the local network (inside network)
- The default firewall behavior is to drop any connections starting from the outside
- Remote hosts typically use protocols such as http, https, ftp, ssh



TCP/IP Ports and Sockets

- On a TCP/IP network every device must have an IP address.
- The **IP address identifies** the **device** e.g. computer.
- However an IP address alone is **not sufficient for running network applications**, as a computer can run multiple applications and/or services.
- Just as the IP address identifies the computer, The **network port identifies** the **application** or **service** running on the computer.
- The diagram below shows a computer to computer connection and identifies the IP addresses and ports.

Cont…



IP Address + Port number = Socket

TCP/IP Ports And Sockets

- A **socket** is the **combination** of **IP address** + **port**
- A **connection between** two **computers uses** a **socket**.

Virtual Private Networks

- A virtual private network (VPN) is a mechanism for creating a secure connection between a computing device and a computer network, or between two networks, using an insecure communication medium such as the public Internet.

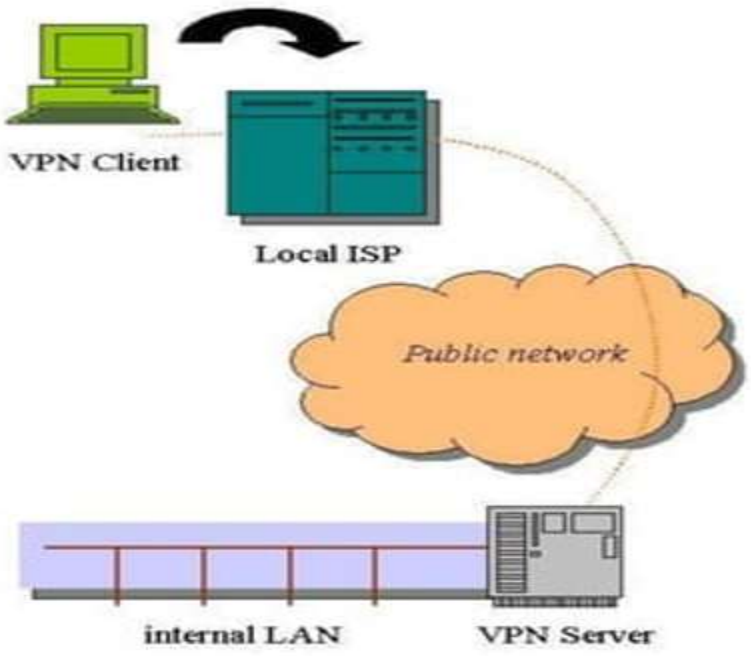
- A VPN can extend access to a private network (one that disallows or restricts public access) to users who do not have direct access to it, such as an office network allowing secure access from off-site over the Internet. The benefits of a VPN include security, reduced costs for dedicated communication lines, and greater flexibility for remote workers.

- A VPN is created by establishing a virtual point-to-point connection through the use of tunneling protocols over existing networks.

Virtual Private Networks

What is VPN

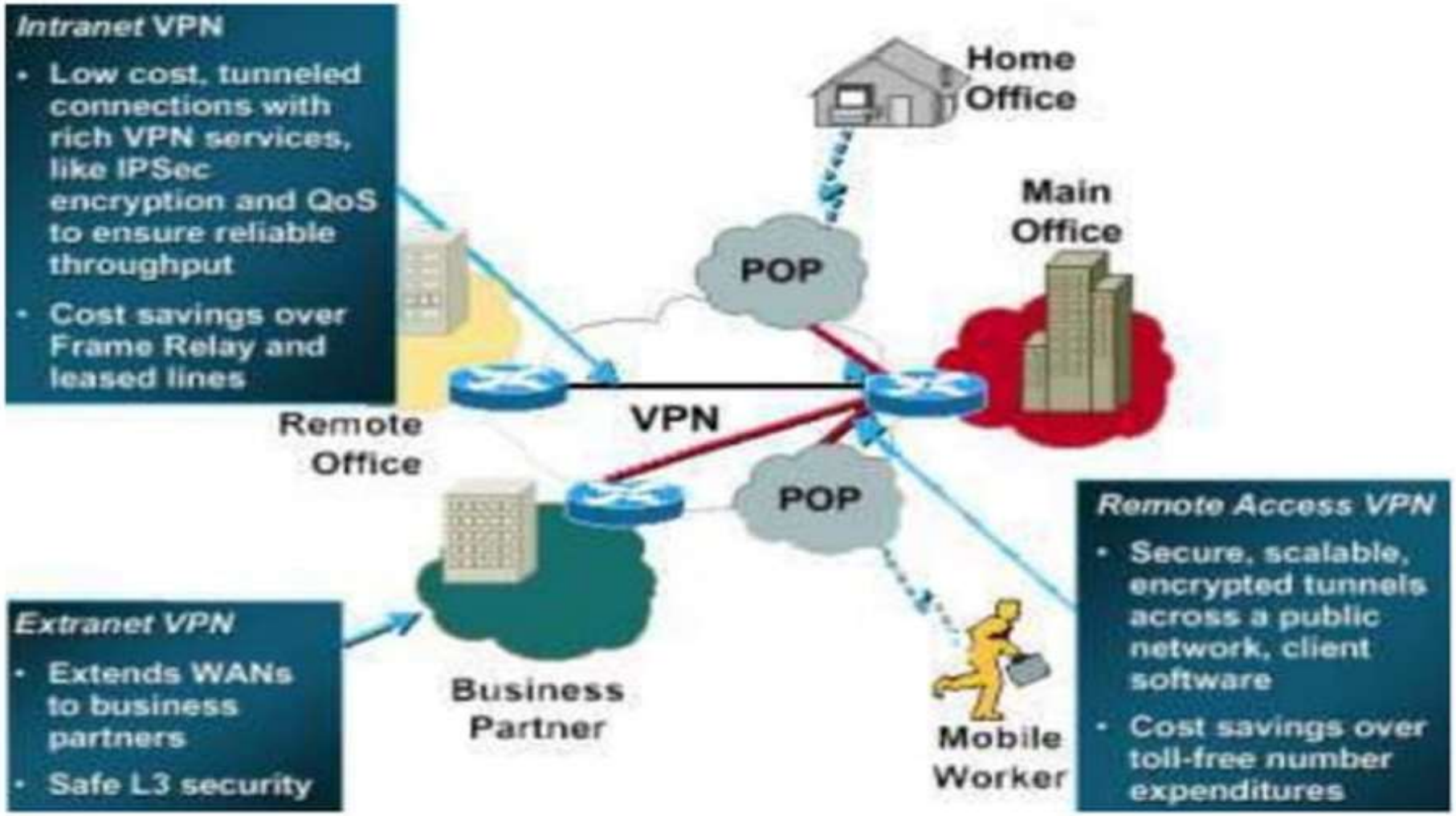
- Virtual Private Network is a type of private network that uses public telecommunication, such as the Internet, instead of leased lines to communicate.
- Became popular as more employees worked in remote locations.
- Terminologies to understand how VPNs work.



Types of VPN

Types of VPN's

- Remote-Access VPN
- Site-to-Site VPN (Intranet-based)
- Site-to-Site VPN (Extranet-based)


Intranet VPN
• Low cost, tunneled connections with rich VPN services, like IPSec encryption and QoS to ensure reliable throughput
• Cost savings over Frame Relay and leased lines
Home Office
Main Office
POP
Remote Office
VPN
POP
Extranet VPN
• Extends WANs to business partners
• Safe L3 security
Business Partner
Mobile Worker
Remote Access VPN
• Secure, scalable, encrypted tunnels across a public network, client software
• Cost savings over toll-free number expenditures

Remote-Access

Remote-Access VPN

- A remote access VPN is for home or travelling users who need to access their central LAN from a remote location.
- They dial their ISP and connect over the internet to the LAN.
- This is made possible by installing a client software program on the remote user's laptop or PC that deals with the encryption and decryption of the VPN traffic between itself and the VPN gateway on the central LAN.

Site-to-site

Site-to-Site VPN

- **Intranet-based** - If a company has one or more remote locations that they wish to join in a single private network, they can create an intranet VPN to connect LAN to LAN.
- **Extranet-based** - When a company has a close relationship with another company (for example, a partner, supplier or customer), they can build an extranet VPN that connects LAN to LAN, and that allows all of the various companies to work in a shared environment.

Snort : Intrusion Prevention System (IPS)

- Snort is an open source network Intrusion Prevention System (IPS) and Intrusion Detection System (IDS).
- It can perform real time traffic analysis and packet-logging on IP networks.
- Also perform protocol analysis, content searching/matching.
- It can be used to detect a variety of attacks, such as buffer overflows, stealth port scans, CGI attacks, SMB probes, OS fingerprinting attempts, and much more.

Cont…

Snort can be configured to run in three modes:

- **Sniffer mode**, which simply reads the packets off of the network and displays them on the screen.
- **Packet Logger mode**, which logs the packets to disk.
- **Network Intrusion Detection System (NIDS) mode**,
- Performs detection and analysis on network traffic.
- The program will monitor network traffic and analyze it against a rule set defined by the user.
- The program will then perform a specific action based on what has been identified.
- With increasing in the growth of internet, it is important task to manage security of network. For this purpose Snort is very useful in term of security.

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